

SECTION 6

Geometry Visualizer

Detailed Student Instructions | Advanced

PROJECT GOAL

Move a slider to update a circle diagram, diameter, circumference, and area.

What you will learn

- Connect index.html, style.css, and script.js.
- Personalize text, colors, and a parent- or teacher-approved picture.
- Modify and test the JavaScript feature for this project.
- Upload the correct folder contents and publish with GitHub Pages.
- Submit both the repository link and the published webpage link.

Files you must keep together

PROJECT ROOT	PURPOSE
index.html	Structure and visible content
style.css	Colors, spacing, and responsive layout
script.js	Interaction and app logic
images/	Profile and project pictures
README.md	Short reference inside the code folder

Repository name used in this guide: **06-geometry-visualizer**

Part A - Open and Personalize the Project

1

Find the correct project folder

Open the downloaded course package and locate **06-geometry-visualizer**. Work on a copy so the original starter remains available.

2

Preview the starter webpage

Double-click **index.html**. It should open in a browser. Keep this browser tab open; after every change, save the file and refresh the tab.

3

Open the code files

Use Visual Studio Code, Notepad, TextEdit in plain-text mode, or another code editor. Open **index.html**, **style.css**, and **script.js**.

Edit index.html

Change the lesson introduction

Explain what radius, diameter, circumference, and area mean in student-friendly language.

Adjust the radius range

The radius input uses min, max, and value. Keep the starting value between the minimum and maximum.

Keep the SVG coordinate system

The viewBox is 0 0 420 420. The JavaScript expects the circle center at x=210 and y=210.

Replace the picture

The starter path is **images/profile-placeholder.svg**. Choose a JPG or PNG that the student has permission to publish. Rename it with lowercase letters and no spaces, place it inside the images folder, and change the path to **images/profile.jpg**.

```

```

Important: If Windows shows profile.jpg.jpg, turn on file extensions and rename it to profile.jpg. GitHub treats uppercase and lowercase filenames as different.

Part B - Customize CSS and JavaScript

1. Change the visual design

Change the diagram fill, circle stroke, and radius-line color. Keep the SVG width at 100% so it remains responsive.

```
:root {
  --bg: #07111f;
  --ink: #f4f7fb;
  --accent: #55d6be;
  --accent2: #ffb45e;
}
```

2. Understand the project logic

Verify the formulas: diameter = $2r$, circumference = $2\pi r$, and area = πr^2 . The toFixed value controls displayed decimal places.

```
diameter = 2 * r
circumference = 2 * Math.PI * r
area = Math.PI * r * r
```

3. Make one meaningful JavaScript change

Changing only a color is not a JavaScript change. Change sample data, text produced by the program, a formula, a setting, a response, or a feature. Save script.js and test the result.

Extension challenge

Add a second mode that calculates a rectangle or triangle.

Privacy and safety check

- Use only approved pictures and general public information.
- Do not publish home addresses, private phone numbers, student IDs, passwords, API keys, or private schedules.
- This starter project does not need an API key. Never paste a secret key into JavaScript.

Part C - Test Before Uploading

Complete every test below. If one test fails, fix it before creating the GitHub repository.

- ☐ Minimum radius produces valid answers.
- ☐ Maximum radius stays inside the SVG box.
- ☐ All rounding choices work.
- ☐ The diagram and numbers update while dragging.

General browser checks

- ☐ index.html opens without a blank page.
- ☐ Text is readable and no content overlaps.
- ☐ The page works when the browser is narrow like a phone.
- ☐ Every picture loads; no broken-image icon appears.
- ☐ Refreshing the page does not create an error.

Confirm the upload structure

```
06-geometry-visualizer/  
|-- index.html  
|-- style.css  
|-- script.js  
|-- README.md  
`-- images/  
    |-- profile.jpg
```

The most common mistake

Do not create an extra nested folder such as **06-geometry-visualizer/06-geometry-visualizer/index.html**. When the repository opens, you should immediately see index.html in the file list.

Choose the final repository name now

Use **06-geometry-visualizer**. Lowercase letters and hyphens make the published address easy to type.

Part D - Upload to GitHub and Publish

1

Sign in

Go to github.com and sign in. Use the account approved for the class.

2

Create a new repository

Select the plus sign in the upper-right area and choose **New repository**. Enter **06-geometry-visualizer** as the repository name.

3

Choose repository settings

Choose **Public** for the simplest GitHub Pages workflow on a free account. Optionally select Add a README file. Click **Create repository**.

4

Open the upload screen

Above the repository file list, choose **Add file**, then **Upload files**.

5

Upload the project contents

Drag `index.html`, `style.css`, `script.js`, `README.md`, and the `images` folder into the upload area. Wait until every filename appears.

6

Commit the upload

Enter the message **Add Geometry Visualizer website**. Commit the changes to the main branch.

7

Open Pages settings

Choose **Settings**. In the left sidebar under Code and automation, choose **Pages**.

8

Select the publishing source

Under Build and deployment, set Source to **Deploy from a branch**. Choose branch **main** and folder **/ (root)**. Click **Save**.

9

Open the site

Return to Settings - Pages after the deployment finishes. Open the displayed site link, or use the address shown below.

<https://YOUR-USERNAME.github.io/06-geometry-visualizer/>

Replace YOUR-USERNAME with the exact GitHub username. Project-page addresses must include the repository name and the final slash.

Part E - Update, Troubleshoot, and Submit

How to update the website later

1

Open the repository

Select the file you need to update, such as index.html, style.css, or script.js.

2

Edit or replace the file

For a text edit, use the pencil icon. For a local replacement, use Add file - Upload files and upload the changed file with the same name.

3

Commit the change

Write a short message describing the change and commit it. GitHub Pages will deploy the updated main branch.

4

Force-refresh the site

If the old version remains, press Ctrl+Shift+R on Windows or Command+Shift+R on Mac.

Troubleshooting

PROBLEM	CHECK
404 page	Pages uses main and / (root); index.html is at the root; URL includes repository name.
No design	Filename is exactly style.css and HTML contains href="style.css".
Buttons fail	Filename is exactly script.js; HTML contains src="script.js"; inspect first console error.
Picture fails	Image exists in images/; spelling, capitals, and extension match the src path.
Site not updated	Confirm the commit is on main; check the Pages deployment; force-refresh.

Final submission checklist

- ☐ Repository link: <https://github.com/USERNAME/REPOSITORY>
- ☐ Published webpage link: <https://USERNAME.github.io/REPOSITORY/>
- ☐ Student name and project section clearly identified for the teacher
- ☐ All required customizations and tests completed

Official GitHub references

[Adding a file to a repository](#)

[Configuring a publishing source for GitHub Pages](#)

[Troubleshooting GitHub Pages 404 errors](#)

[What is GitHub Pages?](#)